Introduction to Heart Anatomy and Physiology

The heart is a muscular pump at the center of the circulatory system, consisting of arteries, veins, and capillaries that transport blood carrying oxygen, nutrients, and waste. Located in the mediastinum beneath the sternum, it measures about the size of a fist (250-350g), shaped like an inverted cone with apex downward/left at 5th/6th rib level and base upward/right at 2nd rib. Surrounded by pericardium (fibrous sac with serous layers: outer fibrous, inner visceral/epicardium; 10-15ml fluid reduces friction), the heart wall has three layers: epicardium (outer serous), myocardium (thick contractile cardiac muscle), endocardium (inner squamous/areolar covering valves/tendons).

This module reviews location/shape, chambers, circulation, valves, coronaries, muscle properties, conduction, and cardiac cycle for foundational cardiac knowledge.

Location, Size, Shape, and Pericardium

Positioned between lungs in mediastinum, heart's apex contacts diaphragm; base attaches major vessels. Pericardium anchors heart, prevents overexpansion, lubricates via fluid in pericardial cavity (between parietal/visceral layers). Epicardium (visceral pericardium) secretes fluid; fibrous pericardium protects/anchors. Heart wall: epicardium (squamous + areolar), myocardium (wrapped cardiac fibers for pumping), endocardium (continuous with vessel endothelium, regulates myocardium).

- Key features: Apex narrow/downward; base broad/upward; ~fist-sized in adults.

Heart Chambers

Four chambers: superior atria (thin-walled, posterior; auricles expand volume), inferior ventricles (thicker; right pumps pulmonary, left systemic). Right/left atria receive blood; ventricles pump via septa. Interventricular septum (thick myocardial wall) separates ventricles; anterior sulcus (diagonal groove) holds anterior interventricular artery/vein/fat; posterior sulcus similar.

Chamber	Location/Features	Function
Right Atrium	Posterior superior; receives vena cavae	Drains systemic deoxygenated blood
Left Atrium	Posterior superior; receives pulmonary veins	Drains oxygenated lung blood
Right		

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pericardium

The word comes from Greek "peri-" meaning around and "kardia" meaning heart. "Pericardium" refers to the sac surrounding the heart, providing protection and anchorage for the organ within the chest.

myocardium

From Greek myo “muscle” and kardia “heart.” Literally “heart muscle.” Names the muscular wall of the heart.

atrium

From Latin atrium “central hall of a Roman house, main room.” Later applied to the heart’s entrance chambers. Implies an “entrance hall” where blood first arrives.

Ventricle Inferior; thinner wall Pumps to pulmonary trunk
Left Ventricle Inferior; thickest wall Pumps to aorta (systemic)
Circulation System
Systemic: superior vena cava (upper body/head/neck), inferior vena cava (thorax/abdomen/legs) right atrium right ventricle pulmonary trunk (branches left/right pulmonary arteries to lungs). Pulmonary: oxygenated blood from lungs via 4 pulmonary veins left atrium left ventricle aorta (systemic).
Major vessels attach at base; ensures unidirectional flow via valves. Blood carries O2/nutrients to organs, CO2 to lungs.
Heart Valve Anatomy
Four valves (fibrous connective tissue + endocardial cover) ensure one-way flow: atrioventricular (AV: tricuspid right/3 cusps, mitral/bicuspid left/2 cusps) between atria/ventricles; semilunar (aortic/pulmonary/3 cusps each) at ventricle outflows. Chordae tendineae (tendons) anchor AV valves to papillary muscles (myocardial cones) preventing prolapse. Valves respond to pressure: low ventricular pressure opens AV (atria fill ventricles); high closes semilunar (backflow pockets); ventricular systole closes AV/opens semilunar.

- Tricuspid: Right AV, 3 flaps.
- Mitral: Left AV, 2 flaps (miter-shaped).
- Pulmonary/Aortic: Semilunar, pocket-shaped cusps.

Coronary Arteries and Veins
Coronary circulation nourishes myocardium: left coronary artery (from aorta above semilunar) anterior interventricular (LAD: anterior ventricles/septum), circumflex (left atrium/ventricle lateral/posterior; marginal branches). Right coronary right marginal (right ventricle lateral), posterior interventricular (PDA: posterior septum; supplies AV node 80%). Variations: right coronary dominant 80-85%; left circumflex 10-15%.

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ventricle
From Latin ventriculus “little belly,” diminutive of venter “belly, abdomen.” Literally “small belly” or “small cavity.” Used for the main pumping chambers of the heart and brain cavities.

aorta
From Greek aortē, traditionally linked to aeirein “to lift, to raise.” Interpreted as “that which is hung up or lifted.” Used for the main arterial trunk leaving the heart.

systole
From Greek systolē “a drawing together, contraction,” from syn “together” and stellein “to send, draw.” Literally “a contracting together.” Refers to the phase when the heart contracts.

tricuspid
From Latin tri- “three” and cuspis “point, cusp.” Literally “three-pointed.” Describes a valve with three pointed cusps or leaflets.

Veins (80% drainage): great cardiac (apex → anterior sulcus → coronary sulcus), small cardiac/middle (posterior), posterior left ventricle → coronary sinus (empties right atrium). Anterior cardiac veins direct to right atrium.

Artery	Path/Supply
Left Anterior Descending	Anterior sulcus; ventricles/septum
Left Circumflex	Coronary sulcus; left atrium/ventricle
Right Coronary	Coronary sulcus; right atrium/ventricle; PDA
Right Marginal	Lateral right ventricle

Cardiac Muscle Tissue Properties

Myocardium: short branched fibers (10-20μm diameter, 50-100μm long; central nucleus; striated A/I bands via myofilaments). Features: numerous/large mitochondria (ATP for contraction), sarcoplasmic reticulum (less organized, no terminal cisternae), T-tubules at Z-discs. Intercalated discs: anchoring (fascia adherens/desmosomes), electrical (gap junctions/connexons) for ion flow, syncytial action (myocardium contracts as unit). Properties: automaticity (self-excitation), conductivity (rapid impulse), contractility (forceful), excitability (depolarization response). Resistant to fatigue via ATP/abundant blood.

Conduction System

Specialized myocardial cells (not neural) initiate/conduct impulses for rhythmic coordinated contraction. Components:

- SA Node: Spindle (10-20mm x 2-3mm), right atrial sulcus terminalis (SVC/RA junction); pacemaker.
- Internodal Pathways: Middle (post-SVC to AV node), posterior (crista terminalis to AV node); atrial myocardium.
- AV Node: Interatrial septum; delays impulse (atria-ventricles sync); arterial supply right coronary (85-90%).
- Bundle of His: Penetrates fibrous body/septum; non-branching.
- Bundle Branches: Left (sheet under aortic cusp), right (intramyocardial to apex).
- Purkinje Fibers: Endocardial network (inner 1/3 ventricles); rapid spread, ischemia-resistant.

Cardiac Cycle

Sequence per beat: diastole (ventricular relaxation/filling), systole (contraction/ejection). Early diastole: semilunar close, AV open, passive fill. Mid: continued fill. Late: atrial contraction (SA impulse). Systole: AV close (S1 sound), semilunar open, ejection (S2 close). Pressure gradient drives flow; normal BP 120/80 mmHg (systolic/diastolic).

Phases:

1. SA fires → atria contract → ventricles fill.
2. AV node delay → bundle/Purkinje → ventricles contract simultaneously.
3. Blood to aorta/pulmonary; cycle repeats (~75 bpm).

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sarcoplasmic reticulum

Sarco- from Greek sarx/sarkos “flesh,” plasma “something molded, formed,” and Latin reticulum “little net.” Literally “little net of the flesh-forming substance.” Names the network of tubules in muscle cell cytoplasm.

purkinje

Eponym from Jan Evangelista Purkyně (Latinized “Purkinje”), the Czech physiologist who described these fibers. The name itself carries no descriptive Latin or Greek meaning in anatomy. It designates the specialized conduction fibers in the heart.

Phase	Events	Valves
Diastole	Ventricles relax/fill	AV open; semilunar closed
Systole	Ventricles contract/eject	AV closed; semilunar open